Consider a plane wall of thickness L, which is to act as shielding for a nuclear reactor. The inner surface receives gamma radiation that is partially absorbed within the shielding and has the effect of an internally distributed heat source. In particular heat is generated per unit volume within the shielding according to the relation

$$q(x) = q_o'' \alpha e^{-\alpha x}$$

Where q_o'' is the incident radiation flux and α is the absorption coefficient of the shielding.

(a) If the inner ($x = 0$) and outer ($x = L$) surfaces of the shielding are maintained at constant temperatures of T_1 and T_2 respectively, find the temperature distribution in the shielding?

(b) Find an expression for the x location in the shield at which the temperature is a maximum.

a) Find T distr. inside the shielding (a distributed heat source inside the wall.

Applicable form of energy eqn at S.S.

$$k \frac{d^2T}{dx^2} + \dot{q} = 0.$$

$$\frac{d^2T}{dx^2} = -\frac{q_o'' \alpha}{k} e^{-\alpha x}$$

$$A = -\frac{q_o'' \alpha}{k}$$

$$\frac{dT}{dx} = \frac{A}{-\alpha} e^{-\alpha x} + C_1$$

$$T = \frac{A}{\alpha^2} e^{-\alpha x} + C_1 x + C_2$$

$$T = -\frac{q_o''}{\alpha k} e^{-\alpha x} + C_1 x + C_2$$

at $x = 0$.

$$T_1 = -\frac{q_o''}{\alpha k} + C_2$$

$$\boxed{C_2 = T_1 + \frac{q_o''}{\alpha k}}$$

$$T - T_1 = \frac{q_o''}{\alpha k}\left(1 - e^{-\alpha x}\right) + C_1 x$$

at $x = L$

$$T_2 - T_1 = \frac{q_o''}{\alpha k}\left(1 - e^{-\alpha L}\right) + C_1 L$$

$$C_1 = \frac{1}{L}\left[(T_2 - T_1) - \frac{q_o''}{\alpha k}\left(1 - e^{-\alpha L}\right)\right]$$

$$\therefore T - T_1 = \frac{q_o''}{\alpha k}\left(1 - e^{-\alpha x}\right) + (T_2 - T_1)\frac{x}{L} - \frac{q_o''}{\alpha k}\left(1 - e^{-\alpha L}\right)\frac{x}{L}$$

$$\boxed{T = T_1 - (T_1 - T_2)\frac{x}{L} + \frac{q_o''}{\alpha k}\left(1 - \frac{x}{L}\right) + \frac{q_o''}{\alpha k}\left(\frac{x}{L} e^{-\alpha L} - e^{-\alpha x}\right)}$$

b) Location at which T is max.

$$\frac{dT}{dx} = \frac{q_0''}{k} e^{-\alpha x} + \frac{(T_2 - T_1)}{L} - \frac{q_0''}{\alpha k L}\left(1 - e^{-\alpha L}\right)$$

$\frac{dT}{dx} = 0$ gives

$$e^{-\alpha x} = \frac{(T_1 - T_2)}{L}\frac{k}{q_0''} + \frac{q_0''}{\alpha k L}\left(1 - e^{-\alpha L}\right)$$

It was decided to melt the ice inside a frozen water pipe by passing an electric current I through the pipe wall. The inner and outer wall of the pipe are r_1 and r_2, and its electrical resistance per unit length is designated as R_e' (Ω/m). The pipe is well insulated on the outside and during melting, the ice (and water) in the pipe remain at a constant temperature T_m associated with the melting process.

(a) Assuming that steady state conditions are reached shortly after application of the current, determine the form of the steady state temperature distribution T(r) in the pipe wall during the melting process.

(b) Develop an expression for the time t_m required to completely melt the ice. Calculate the time for I = 100 A, $R_e' = 0.30$ (Ω/m) and r_1 = 50 mm.

a) The appropriate form of the heat eqn and the general soln is

$$T(r) = -\frac{\dot{q}}{4k} r^2 + C_1 \ln r + C_2, \quad \text{where } \dot{q} = \frac{I^2 Re}{\pi(r_2^2 - r_1^2)}$$

$$\left[\frac{1}{r}\frac{d}{dr}\left(r\frac{dT}{dr}\right) + \frac{\dot{q}}{k} = 0.\right] *$$

BC 1 $\quad \left.\frac{dT}{dr}\right|_{r=r_2} = 0 \Rightarrow C_1 = \frac{\dot{q} r_2^2}{2k}$.

$$\therefore \quad T(r) = -\frac{\dot{q}}{4k} r^2 + \frac{\dot{q} r_2^2}{2k} \ln r + C_1$$

BC 2 $\quad T(r_1) = T_m. \Rightarrow T_m = -\frac{\dot{q}}{4k} r_1^2 + \frac{\dot{q} r_2^2}{2k} \ln r_1 + C_1$

$$\Rightarrow T(r) = T_m + \frac{\dot{q} r_2^2}{2k} \ln\frac{r}{r_1} - \frac{\dot{q}}{4k}(r^2 - r_1^2).$$

b) Conservation of energy dictates that the energy reqd. to completely melt the ice, Em, must equal the energy which reaches the inner surface of the pipe by conduction through the wall

$$\Delta E_{st} = t_m \, q_{cond\, r_1}$$

For a unit length of the pipe

$$\rho \pi r_1^2 \cdot h_{sf} = t_m \left[k \, 2\pi r_1 \left.\frac{dT}{dr}\right|_{r_1}\right]$$

$$\rho \pi r_1^2 h_{sf} = 2\pi r_1 k t_m \left[\frac{\dot{q} r_2^2}{2k r_1} - \frac{\dot{q} r_1}{2k}\right].$$

$$\rho \pi r_1^2 h_{sf} = t_m \dot{q} \pi (r_2^2 - r_1^2) = t_m I^2 Re$$

$$t_m = \frac{\rho h_{sf} r_1^2}{\dot{q}(r_2^2 - r_1^2)} = \frac{\rho h_{sf} \pi r_1^2}{I^2 Re'}$$

with $r_1 = 0.05$ m, I = 100 A and $Re' = 0.3\ \Omega$/m.

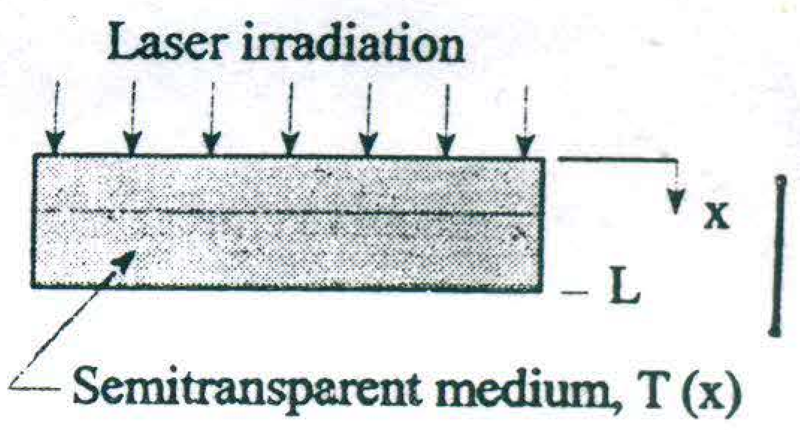
$$t_m = \frac{920 \times 3.34 \times 10^5 \times \pi \times (0.05)^2}{(100)^2 \times 0.3}$$

$$\boxed{t_m = 804 s.}$$

$$* \quad r\frac{dT}{dr} = -\frac{\dot{q} r^2}{2k} + C_1$$

$$T = -\frac{\dot{q}}{4k} r^2 + C_1 \ln r + C_2.$$

The steady state temperature distribution in a semi-transparent material of thermal conductivity k and thickness L exposed to laser irradiation is of the form, $T(x) = -(A/ka^2)\,e^{-ax} + Bx + C$, where A, a, B and C are known constants. For this situation, radiation absorption in the material is manifested by a distributed heat generation term, $\dot{q}(x)$.



(a) Obtain expressions for the conduction heat fluxes at the front and rear surfaces.

(b) Derive an expression for $\dot{q}(x)$.

(c) Derive an expression for the rate at which radiation is absorbed in the entire material, per unit surface area. Express your result in terms of the known constants for the temperature distribution, the thermal conductivity of the material, and its thickness.

(a) The surface heat fluxes to be obtained by Fourier's law

$$q''_x = -k\frac{dT}{dx} = -k\left[-\frac{A}{ka^2}\cdot(-a)e^{-ax} + B\right]$$

Front surface, $x=0 \Rightarrow q''_x(0) = -\left[\frac{A}{a} + kB\right]$

Rear surface, $x=L \Rightarrow q''_x(L) = -\left[\frac{A}{a}e^{-aL} + kB\right]$.

(b) The heat diffⁿ eqⁿ for the medium is

$$\frac{d}{dx}\left(\frac{dT}{dx}\right) + \frac{\dot{q}}{k} = 0.$$

$$\therefore\ \dot{q}(x) = -k\frac{d^2T}{dx^2} = -k\frac{d}{dx}\left[\frac{A}{ka}e^{-ax} + B\right] = Ae^{-ax}.$$

(c) Performing an energy balance on the medium

$$\dot{E}_{in} - \dot{E}_{out} + \dot{E}_g = 0.$$

$\dot{E}_g$ represents the absorbed radiation. On a unit area basis

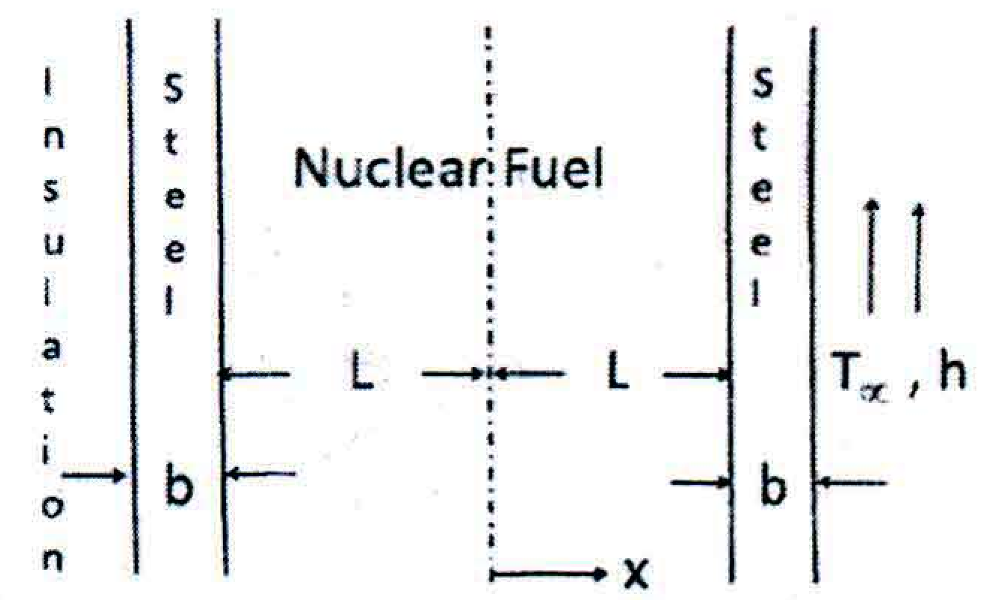
$$\dot{E}''_g = -\dot{E}_{in} + \dot{E}_{out} = -q''_x(0) + q''_x(L)$$

$$\boxed{\dot{E}''_g = \frac{A}{a}\left(1-e^{-aL}\right)}$$

OR

$$\dot{E}''_g = \int_0^L \dot{q}(x)\,dx = \int_0^L Ae^{-ax}\,dx = \frac{A}{a}\left(1-e^{-aL}\right)$$

A nuclear fuel element of thickness 2L is covered with a steel cladding of thickness b on both sides as shown in the adjoining figure. Heat generated in the nuclear fuel at a rate of q (W/m^3) is removed by a fluid at T_∞, which adjoins one surface and is characterized by a convection coefficient h. The other surface is well insulated, and the fuel and the steel have thermal conductivities k_f and k_s respectively. Obtain an expression for the temperature distribution T(x) in the nuclear fuel at steady state. Express your answers in terms of q, L, b, k_f, k_s, h and T_∞. Sketch the temperature distribution for the entire system.



1-D, SS, Uniform generation.

For $-L \leq x \leq +L$.

$$\frac{d^2T}{dx^2} + \frac{\dot q}{k_f} = 0.$$

$$T = -\frac{\dot q}{2k_f}x^2 + C_1 x + C_2.$$

Insulated wall at $x = -(L+b)$

$\therefore$ Heat flux at $x = -L$ is zero (at SS, no heat can go to the steel).

$$\therefore \left.\frac{dT}{dx}\right|_{x=-L} = -\frac{\dot q}{\cancel{2}k_f}(-L) + C_1 = 0 \Rightarrow C_1 = -\frac{\dot q L}{k_f}.$$

$$\therefore T = -\frac{\dot q}{2k_f}x^2 - \frac{\dot q L}{k_f}x + C_2. \quad (1)$$

For the steel cladding on the right

$$\underset{\text{(in fuel)}}{\dot E_g} = \underset{\text{(through steel)}}{q_{cond.}} = q_{conv.} \quad (2)$$

$$\dot q(2L) = \frac{k_s}{b}(T_{s_1} - T_{s_2}) = h(T_{s_2} - T_\infty) \quad (3)$$

$$\therefore T_{s_1} = \cancel{\frac{\dot q}{2Lb}}\ \frac{\dot q(2Lb)}{k_s} + T_{s2} \quad (4)$$

$$\therefore \quad T_{s_1} = \frac{\dot{q}_v (2Lb)}{k_s} + \frac{\dot{q}_v (2L)}{h} + T_\infty \quad (5)$$

~~Thus~~

From (1)

$$T_{s_1} \left(= T \text{ at } x = L\right) = \frac{\dot{q}_v (2Lb)}{k_s} + \frac{\dot{q}_v (2L)}{h} + T_\infty$$

$$= -\frac{3}{2} \frac{q_v (L^2)}{k_f} + C_2$$

$$\therefore C_2 = T_\infty + q_v L \left[\frac{2b}{k_s} + \frac{2}{h} + \frac{3}{2} \frac{L}{k_f} \right].$$

Thus the temperature distribution is

$$T = -\frac{\dot{q}_v}{2k_f} x^2 - \frac{\dot{q}_v L}{k_f} x + q_v L \left[\frac{2b}{k_s} + \frac{2}{h} + \frac{3}{2} \frac{L}{k_f} \right] + T_\infty$$